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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.

Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

Step 1: Gathering Data

To construct the portfolio we considered data from 2021-01-04 to 2024-12-31. Then, to evaluate the performance of the portfolio we are going to use the data from 2025-01-06 to 2025-05-08.

We downloaded the stocks for the following S&P 500 tickers from yahoo finance:

Sectors/company					
Materials	LIN (Linde plc)	APD (Air Products and Chemicals Inc.)	ALB (Albemarle Corporation)		
Communication Services	META (Meta Platforms Inc.)	CMCSA (Comcast Corporation)	FOXA (Fox Corporation)	GOOGL (Alphabet Inc.)	
Energy	XOM (Exxon Mobil)	COP (ConocoPhillips)	APA (APA Corporation)	CVX (Chevron Corporation)	
Financials	JPM (JPMorgan Chase & Co)	PNC (PNC Financial Services)	LNC (Lincoln National)	BRK-B (Berkshire Hathaway)	V (Visa Inc.)
Industrials	CAT (Caterpillar Inc.)	UPS (United Parcel Service)	CAPR (Carrier Global Corporation)	GE (General Electric)	
Information Technology	NVDA (NVIDIA Corporation)	SNPS (Synopsys Inc.)	ZS (Zscaler Inc.)	AAPL (Apple Inc.)	MSFT (Microsoft)
Consumer Staples	PG (Procter & Gamble Co.)	KMB (Kimberly-Clark Corp.)	TAP (Molson Coors Beverage)	COST (Costco Wholesale Corp.)	
Utilities	NEE (NextEra Energy Inc.)	DUK (Duke Energy Corp.)	AES (AES Corp.)		
Health Care	UNH (UnitedHealth Group)	MDT (Medtronic plc)	BAX (Baxter International)	LLY (Eli Lilly)	
Consumer Discretionary	AMZN (Amazon Inc)	NKE (NIKE Inc.)	ETSY (Etsy Inc.)	TSLA (Tesla Inc.)	

Real Estate	PLD (Prologis Inc.)	AMT (American Tower Corp)	AVB (AvalonBay Communities)		
Extra Assets	GC=F (Gold)	BTC-USD (Bitcoin)			

- We used the Close feature column and calculate the geometric returns as formula:

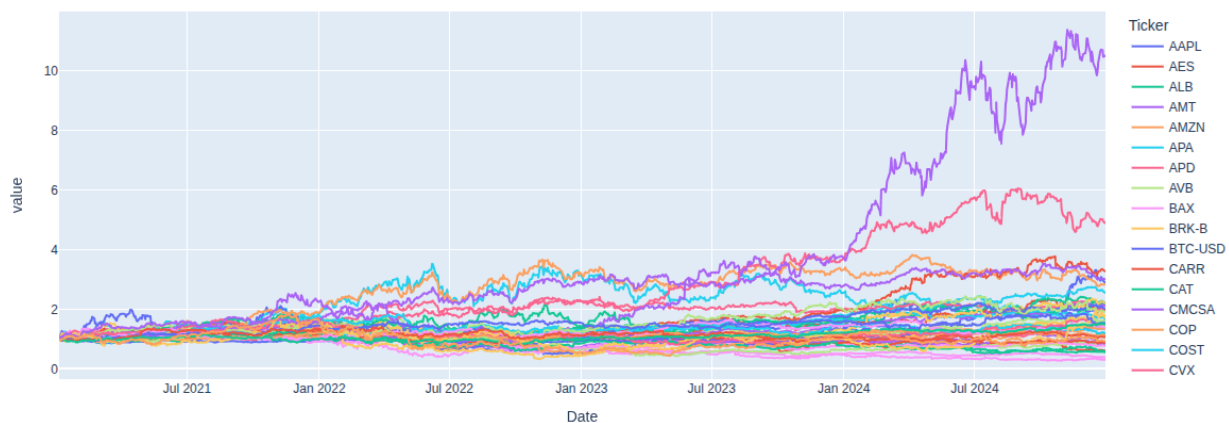
$$rt = \log(P_t / P_{t-1})$$

then we calculated mean returns.

- We downloaded daily 2-year Treasury yield from the FRED database (DGS2 series) as Risk Free Rate of return. Converted annual rate to daily risk free rate by dividing by trading days (as 252). Then we calculated the mean daily risk free rate.

Normalized Stock prices growth Plots for Train and Test data:

Train prices



Test prices



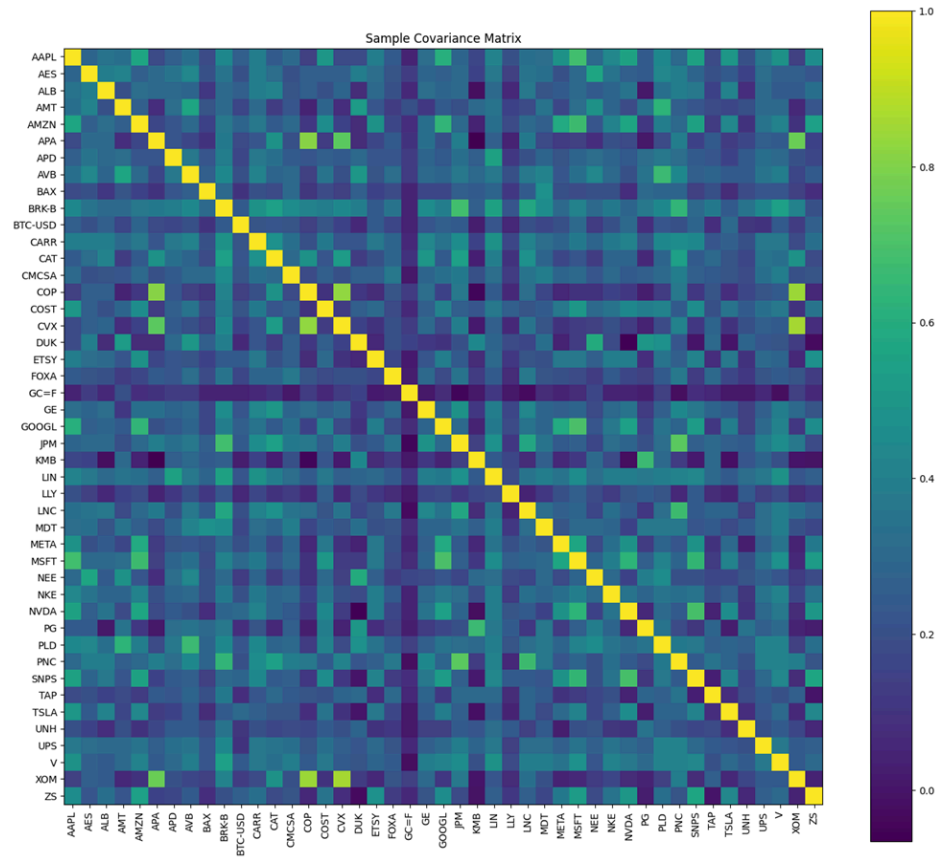
It can be seen that in the historical data used to construct the portfolio there are some assets that are outperforming the majority. But for the testing period we don't observe this discrepancy. In fact, the assets that are outperforming the others aren't the same as the training period.

To estimate the covariance of returns we used two approaches — one using the sample estimator, and one using the Ledoit–Wolf shrinkage estimator.

- The Sigma_sample is a matrix where diagonal elements are variance of each asset and off-diagonal represents covariance between pairs of assets.
- The Ledoit–Wolf shrinkage estimator is more stable and is recommended for practical applications, specially for large portfolios.

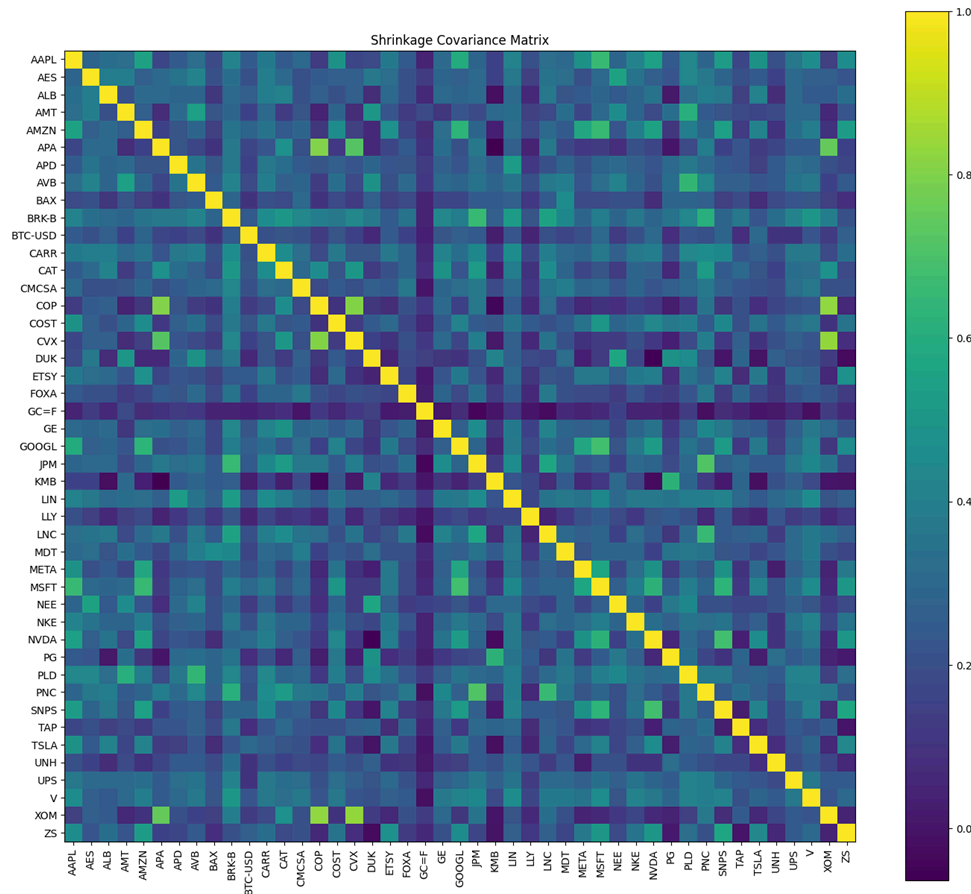
First lines of Sigma_Sample Estimator:

Sigma_sample													
LIN	0.000096	0.000109	0.000139	0.000076	0.000100	0.000111	0.000121	0.000069	0.000066	0.000080	...	0.000092	0.000100
LLY	0.000065	0.000058	0.000037	0.000045	0.000075	0.000026	0.000036	0.000051	0.000037	0.000045	...	0.000066	0.000019
LNC	0.000159	0.000219	0.000341	0.000089	0.000199	0.000339	0.000099	0.000150	0.000127	0.000177	...	0.000155	0.000342
MDT	0.000072	0.000097	0.000097	0.000077	0.000079	0.000074	0.000064	0.000079	0.000112	0.000067	...	0.000086	0.000088
META	0.000235	0.000126	0.000233	0.000082	0.000371	0.000106	0.000101	0.000091	0.000097	0.000093	...	0.000135	0.000113
MSFT	0.000187	0.000109	0.000161	0.000087	0.000246	0.000072	0.000077	0.000076	0.000055	0.000071	...	0.000113	0.000083
NEE	0.000086	0.000208	0.000133	0.000136	0.000088	0.000045	0.000090	0.000101	0.000061	0.000056	...	0.000131	0.000060
NKE	0.000153	0.000159	0.000198	0.000099	0.000191	0.000133	0.000097	0.000088	0.000073	0.000085	...	0.000125	0.000123



First lines of Sigma_shrinkage Estimator:

Sigma_shrinkage													
LIN	0.000094	0.000107	0.000136	0.000075	0.000098	0.000109	0.000119	0.000068	0.000065	0.000078	...	0.000090	0.000098
LLY	0.000064	0.000057	0.000036	0.000044	0.000074	0.000025	0.000035	0.000050	0.000037	0.000044	...	0.000065	0.000019
LNC	0.000157	0.000215	0.000335	0.000087	0.000196	0.000333	0.000097	0.000147	0.000125	0.000174	...	0.000152	0.000336
MDT	0.000071	0.000095	0.000096	0.000076	0.000078	0.000073	0.000063	0.000078	0.000110	0.000066	...	0.000085	0.000086
META	0.000231	0.000124	0.000229	0.000080	0.000364	0.000105	0.000099	0.000089	0.000096	0.000091	...	0.000133	0.000111
MSFT	0.000184	0.000107	0.000158	0.000085	0.000241	0.000070	0.000075	0.000075	0.000054	0.000070	...	0.000111	0.000081
NEE	0.000084	0.000204	0.000131	0.000134	0.000086	0.000044	0.000088	0.000099	0.000059	0.000055	...	0.000128	0.000059
NKE	0.000150	0.000157	0.000194	0.000098	0.000187	0.000131	0.000095	0.000086	0.000072	0.000083	...	0.000123	0.000121



Step 2: Markowitz Optimization

To determine the weights of the assets in the portfolio we employed the classical mean-variance model. In our portfolio we don't allow for shorting and also the maximum weight of an asset can be 0.35. In the optimization process we examined different upper bounds from 0.15 to 0.40. The performance of the model is highly affected by this threshold. Lower values of threshold helps in a better diversification but not necessarily better performance of the portfolio during the selected testing time.

The function implements mean-variance optimization as condition trying to find the maximum Sharpe ratio portfolio, assuming:

- A **long-only constraint** (no short selling),
- Optional **upper bounds** on asset weights,
- Use of the **SLSQP optimizer** from `scipy.optimize`.

where

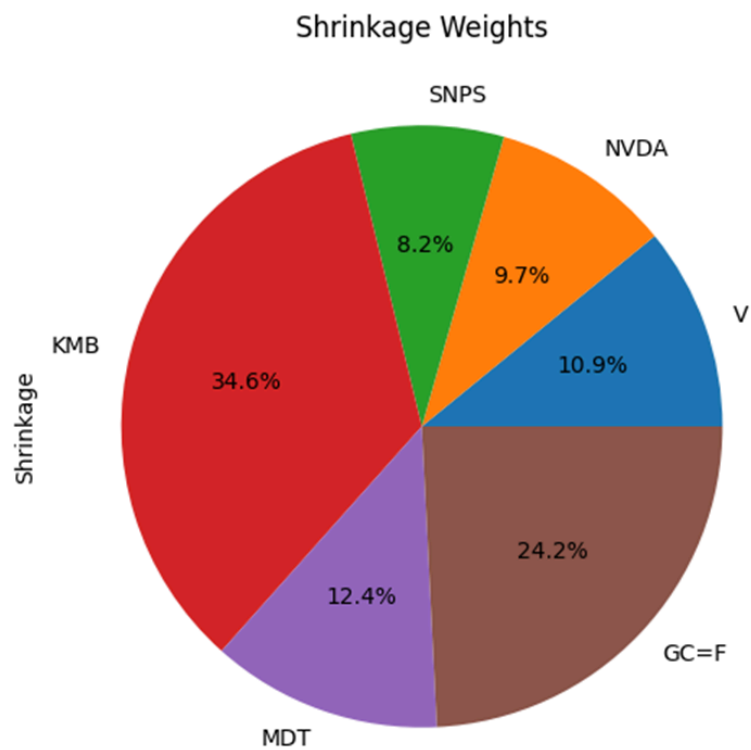
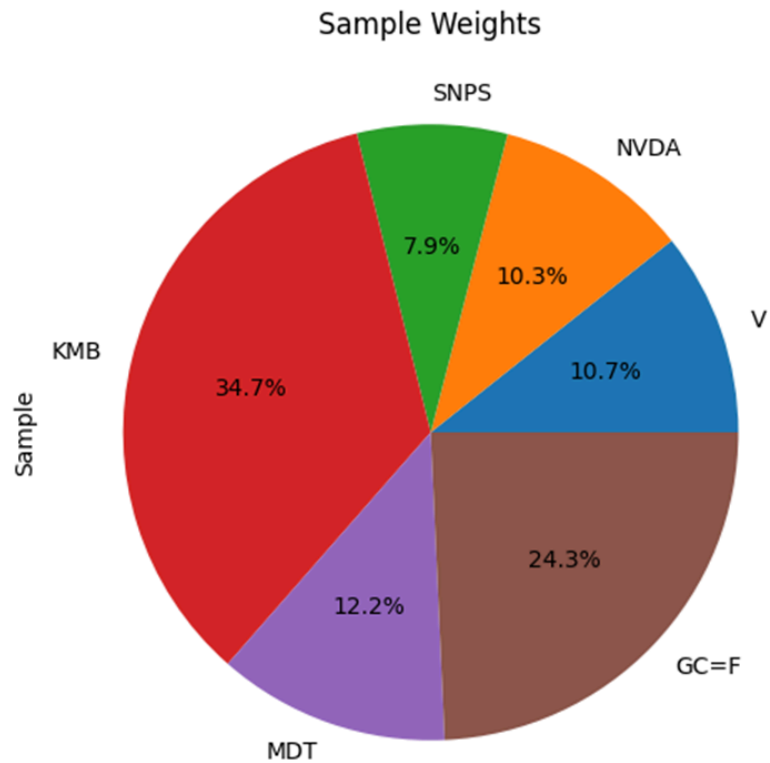
- w : Portfolio weights (vector)

- μ : Expected daily returns (vector)
- Σ : Covariance matrix of returns
- r_f : Daily risk-free rate
- $\text{portf_mu} = w @ \mu - r_f$ # daily excess return
- $\text{portf_sig} = \text{np.sqrt}(w @ \Sigma @ w)$
- $\text{portf_sharpe_ratio} = \text{port_mu} / \text{port_sig}$
- constraints:
 - bounds: Each asset weight is between 0 and w_{max} (default 1.0 $\rightarrow$ no leverage).
 - cons: A constraint that ensures the portfolio is fully invested (weights sum to 1).
 - w0: Starting point for optimization (equal weighting).

The resulting weights for two approaches of calculating the covariance are as follows. It is worthwhile to mention that, here we are displaying only the assets that were selected in the optimization process.



	Sample	Shrinkage
V	0.107	0.109
NVDA	0.103	0.097
SNPS	0.079	0.082
KMB	0.347	0.346
MDT	0.122	0.124
GC=F	0.243	0.242



It can be seen that in both approaches the composition of the portfolio is the same and the weights are very close. The reason for this similarity is that our portfolio is not too big and we have enough data points compared to the number of the securities.

It's time to evaluate the performance of the portfolio during the testing period. To understand the situation better we are going to compare the performance for both train and test periods.

Portfolio stats:				
	Sample Σ		Shrink Σ	
	Train	Test	Train	Test
μ_{ann}	0.074	0.181	0.071	0.185
σ_{ann}	0.130	0.176	0.129	0.175
Sharpe	0.572	1.029	0.553	1.059
TotalReturn	0.469	0.070	0.451	0.072

As expected both portfolios --constructed using different approaches to estimating the covariance-- have a very similar performance. There some points to discuss here

- The Sharpe ratio in the training process is less than one. It usually is interpreted that the portfolio composition is not good and the excess risk is not compensated by enough return.
- The volatility in the testing period is substantially higher than the training period. It shows that the market is passing through an unstable high fluctuations phase.
- The total return and also Sharpe ratio are negative for the evaluation (test) period. So, we are losing money!
- Looking at the news we can see why we are facing this situation. In our interpretation, the global financial markets, not only the US, is experiencing a high volatility time due to new tariff policies of the US government.

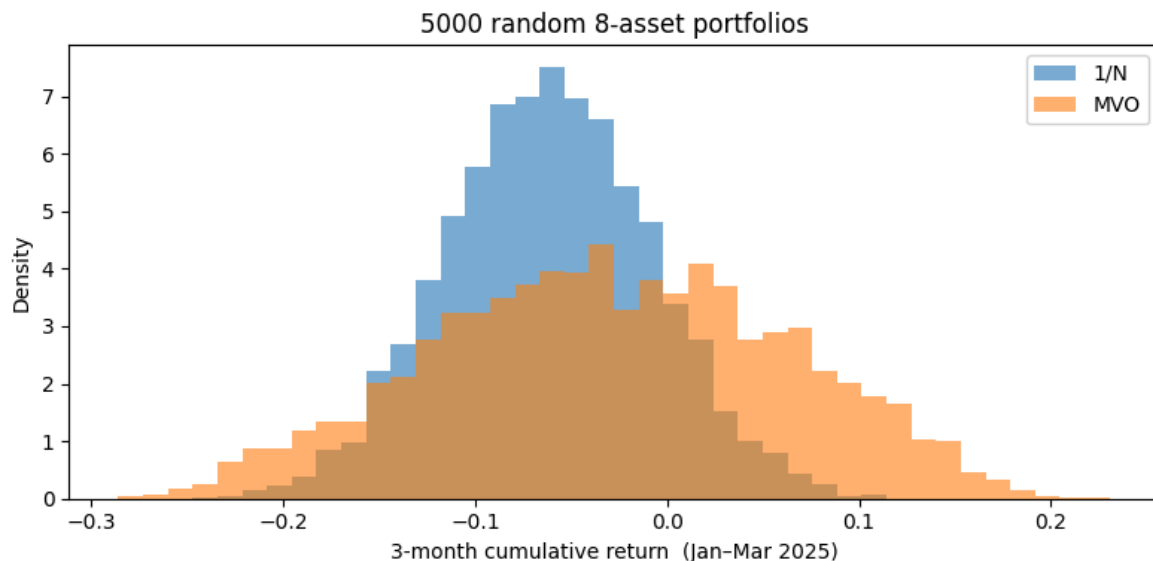
Risk contribution table (train Σ):

	Sample Σ	Shrink Σ
Asset		
GC=F	9.0%	9.5%
KMB	27.6%	28.7%
MDT	10.8%	11.1%
NVDA	27.5%	25.1%
SNPS	13.8%	14.2%
V	11.3%	11.5%

We can see the highest risk contribution by KMB, NVDA followed by V and SNPS.

Step 3: Random Strategy Optimization

To see if we could have a better performance, let's compare two situations. The strategy is as follows. We randomly choose 8 assets from our universe and construct an equal-weights (1/N) portfolio. Then we compare the performance of this portfolio with a portfolio in which its weights are calibrated by mean-variance optimization (MVO). The performance evaluation is done for the testing period. We did this process 5000 times and the result is displayed in the below graph. We used MonteCarlo simulation to allocate equally across 8 randomly selected portfolios iteratively and then calculate cumulative returns.



From the plot, it seems that the portfolios based on the MVO performed better than equally-weighted ones.

Step 4: Black-Litterman

First, we must declare our views as the portfolio manager to obtain the prior values to feed into the Black-Litterman (BL) model. We did a search into the technical news and the report on the dates before 2024/12/31 to gather relevant news about the firms in our universe.

Views

Material Sector

- Linde (LIN) – Absolute view. Q: $\sim +10\%$. Confidence: Medium. Justification: The company underperformed in 2024 but it is expected to improve in 2025, if the global economy improves. CNBC (Dec 27, 2024) noted Linde could return to its “beat-and-raise” earnings cadence with a stronger economy. The company ended Q3 2024 with a high record of backlogs that is a positive factor for growth. [<https://www.cnbc.com/2024/12/27/a-pickup-in-the-global-economy-should-get-linde-back-to-its-winning-ways-in-2025.html>]
- Air Products & Chemicals (APD) – Absolute view. Q: $\sim +15\%$. Confidence: Medium. Justification: An activist investor called Mantle Ridge took more than \$1B stake in APD in 2024, pushing for strategic changes (succession planning, capital allocation) to unlock value. The stock increased about 9% on the news in Oct 2024. With these catalysts, analysts expect improved efficiency and shareholder returns. [<https://www.investopedia.com/air-products-stock-jumps-as-activist-investor-pushes-strategy-shift-8724566>]
- Albemarle (ALB) – Absolute view. Q: $\sim +15\%$. Confidence: Medium (high upside, but commodity-driven). Justification: ALB’s shares were depressed in 2024 due to lower global lithium prices, but analysts anticipate a strong recovery due to the increasing demand for batteries. The average analyst target implies $\sim 15\text{--}20\%$ rise from late-2024 prices. [<https://www.zacks.com/stock/quote/ALB>]

Communication Services

- Meta Platforms (META) – Absolute view. Q: $\sim +15\%$. Confidence: High. Justification: Meta’s AI and advertising momentum are creating bullish forecasts. Multiple agencies have raised targets in Dec 2024 (e.g. JPMorgan to \$725, Truist \$700, Bernstein \$675) as Meta’s earnings and user growth exceeded peers. Consensus calls for $\sim 10\text{--}20\%$ upside as Meta’s ad revenue and new AI features drive growth.

<https://investorsobserver.com/news/meta-platforms-meta-stock-prediction-for-2025-betting-on-ai>)

- Comcast (CMCSA) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: Comcast didn't have good results in 2024 (stock -13.6% vs S&P +25%), but for 2025 analysts are moderately optimistic. The stocks are traded below the valuation target, and consensus rating was Moderate Buy (17 Buy, 11 Hold). Strong growth and cost controls support a modest rebound.
(<https://www.nasdaq.com/articles/comcasts-q4-2024-earnings-what-expect>)
- Fox Corp (FOXA) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: Analysts' consensus rating on Fox Corporation's stock is cautiously optimistic, with a "Moderate Buy" overall rating. Out of 20 analysts covering the stock, opinions include nine "Strong Buys" and 11 "Holds." The average analyst price target for FOXA is \$47.65, which indicates that the stock trades at a premium.
(<https://www.nasdaq.com/articles/what-expect-fox-corporations-q2-2025-earnings-report>)
- Alphabet/Google (GOOGL) – Absolute view. Q: ~+10%. Confidence: High. Justification: Google enjoyed the bull market of tech stocks in 2024 and the analysts expect a similar performance in 2025. Many studies show that the growth indicators are strong. Google is expected to benefit from the AI boom and the strong economy.
(<https://www.investors.com/news/technology/google-stock-buy-now-alphabet-stock-december-2024/>)
- Relative view – AAPL vs. MSFT: Microsoft is expected to outperform Apple. Q: +~7% (MSFT over AAPL). Confidence: High. Justification: Microsoft's revenue/EPS is projected to grow about 15% in 2024–25 vs. Apple's ~6–8%. With its cloud infrastructure & OpenAI advantages and the Activision acquisition, MSFT has a stronger catalysts. One analysis explicitly argues that the MSFT will outperform the AAPL in 2025. Meanwhile Apple, though stable, doesn't have near-term drivers beyond usual services growth.
(<https://www.nasdaq.com/articles/will-microsoft-still-be-worth-more-than-apple-by-2025>), <https://schwabnetwork.com/video/2024-year-in-review-a-i-s-role-in-aapl-msft-nvda->)

Energy Sector

- Exxon Mobil (XOM) – Absolute view. Q: ~+15%. Confidence: High. Justification: Despite a soft 2024, Exxon is expected to have a 'stronger year than 2024' in 2025, per analysts. Free cash flow remains very high (> \$30B/yr), and a pro-fossil U.S. policy shift (new administration in 2025) could boost margins via deregulation. Consensus price targets around \$124 imply ~16% upside.
(<https://www.nasdaq.com/articles/3-reasons-why-exxon-xom-set-excellence-2025>)
- ConocoPhillips (COP) – Absolute view. Q: ~+10%. Confidence: Medium. Justification: COP shares are near the 52-week low, impacted by the supply chain issues. Despite the

operational strength and increased production level, the analysts expect the challenging commodity price environment to persist into 2025. (<https://seekingalpha.com/article/4746461-conocophillips-attractive-even-in-a-more-mute-d-commodity-price-environment>)

- APA Corp (APA) – Absolute view. Q: ~-5%. Confidence: Medium-Low. Justification: In the APA third quarter 2024 the revenue increased by 10% from 3Q 2023 and the net loss reduced by 149% from 3Q 2023. The analysts don't expect that the company will improve in 2025 very much despite the strong production level. (<https://finance.yahoo.com/news/apa-third-quarter-2024-earnings-121345607.html>)
- Chevron (CVX) – Absolute view. Q: ~-5%. Confidence: Medium. Justification: Chevron faces significant risks. Oil price volatility remains a key concern, as lower prices due to the global economy situation could affect the profit of the company. Geopolitical risks, particularly in regions like Kazakhstan, could also disrupt operations. (<https://www.nasdaq.com/articles/chevrons-spending-shift-should-you-buy-stock-2025>)

Financials

- JPMorgan Chase (JPM) – Absolute view. Q: ~+7%. Confidence: Medium. Justification: After a ~40% stock uprising in 2024, it is expected to have modest gains. JPM's profits are at record highs (Q4 2024 net income +30% YoY) and it consistently beat estimates, but with its valuation now higher, we can not expect a high return for 2025. (<https://www.globenewswire.com/news-release/2024/11/18/2982958/28124/en/2024-J-P-Morgan-Private-Bank-Competitor-Profile-Key-Strategies-Financial-Performance-Customers-and-Products-Brand-Building-Activities.html>, <https://www.investopedia.com/what-analysts-think-of-jpmorgan-chase-stock-ahead-of-earnings-8771652>)
- PNC Financial (PNC) – Absolute view. Q: ~+7%. Confidence: Medium. Justification: The banking industry in general has been preparing for a potential fallout from the higher-for-longer interest rate environment by increasing its loan loss provisions, anticipating that some borrowers may have difficulties repaying their debts. Despite these challenges, PNC is expected to have modest gains in 2025. (<https://www.investopedia.com/what-analysts-think-of-jpmorgan-chase-stock-ahead-of-earnings-8771652>)
- Lincoln National (LNC) – Absolute view. Q: ~+5%. Confidence: Medium-Low. Justification: In the Q3 2024, Lincoln National experienced a net loss of \$528 million compared to the Q3 2023 net income of \$853 million. The Life Insurance segment reported an operating income of \$22 million against the previous year quarter's loss of \$173 million. We have a mix of different signals. We remain cautiously optimistic about the company's prospects for 2025, but we expect modest gains. (<https://finance.yahoo.com/news/lincoln-national-q3-earnings-beat-184700902.html>,

<https://www.businesswire.com/news/home/20241101089053/en/Lincoln-National-Corporations-Board-of-Directors-Declares-Series-D-Preferred-Stock-Dividend>)

- Berkshire Hathaway (BRK.B) - Absolute view. Q: ~+10%. Confidence: High. Justification: Berkshire Hathaway made some big moves in 2024 -- it was mainly increasing its cash. The company bought a few new stocks but not in a significant amount. It seems that Buffett is not optimistic about the market and is waiting for a bear market. This strategic decision is expected to pay off in a long horizon. We would like to keep the stock in the portfolio for the mid-long term and expect even in 2025 we will see good news.
(<https://www.fool.com/investing/2024/12/08/cash-and-patience-berkshires-2024-moves-set-the/>)
- Visa (V) - Absolute view. Q: +10%. Confidence: High. Justification: Visa's growth trend remains intact. The consensus rating is Strong Buy. There is no reason why the company will not continue making profit. For the 2024 results we have a revenue of US\$35.9B (up 10.0% from FY 2023) and net income: US\$19.7b (up 16% from FY 2023) and profit margin: 55% (up from 52% in FY 2023). The increase in margin was driven by higher revenue. (<https://finance.yahoo.com/news/visa-full-2024-earnings-line-130723645.html>)

Industrials

- Caterpillar (CAT) – Absolute view. Q: -5% (slight negative). Confidence: Medium. Justification: The macroeconomic conditions are challenging for heavy industry. Despite Caterpillar's strong production level, the analysts expect the challenging commodity price environment to persist into 2025.
(<https://finance.yahoo.com/news/caterpillar-cat-held-better-peers-141032055.html>,
<https://www.forbes.com/sites/greatspeculations/2024/10/28/should-you-pick-caterpillar-stock-ahead-of-q3/>)
- UPS (UPS) – Absolute view. Q: -5%. Confidence: Medium. Justification: UPS has a good strategy for a long term, but the company now is behind his main competitor FedEx. The industry is under pressure in an oversupplied package delivery network. The pandemic threw a wrench into forecasting, and the industry overexpanded routes and capabilities in anticipation of sustained growth that never materialized. Throw in supply chain and inflation challenges in recent years, and it has been far from an easy time for these companies.
(<https://www.nasdaq.com/articles/fedex-outperformed-ups-2024-which-dividend-stock-better-buy-2025>)
- Carrier Global (CARR) – Absolute view. Q: ~+5%. Confidence: High. Justification: HVAC giant Carrier Global reported worse-than-expected sales as the company moved to exit its Fire & Security business. This report causes a drop in the stock price. This movement is still persistent but due to high quality portfolio transformation of the

company and the strong revenue we believe that the company will change the situation in the 2025 and the stock price will go up.
(<https://www.investopedia.com/carrier-global-stock-cools-on-soft-sales-guidance-cut-8733534>)

- General Electric (GE) – Absolute view. Q: ~+15%. Confidence: High. Justification: GE's Aerospace is now the core of GE and it's growing fast. The company expects a double-digit revenue and profit for 2025. All the indicators are pointing to a strong performance in 2025.
(https://www.geaerospace.com/sites/default/files/geaerospace_webcast_pressrelease_01232025.pdf)

Technology

- NVIDIA (NVDA) – Absolute view. Q: +20%. Confidence: Medium. Justification: NVIDIA had an explosive 2024 (AI chip demand). The demand for GPUs is still very high and we don't see any real competitors in the market. Nvidia's data center revenue rose by a remarkable 112% year over year in the third quarter of its fiscal 2025 to a record \\\$30.8 billion, while AMD's revenue from this segment came in at just \\\$3.5 billion in its most recently reported quarter.
(<https://finance.yahoo.com/news/prediction-artificial-intelligence-ai-chip-130700235.html>)
- Synopsys (SNPS) – Absolute view. Q: -5%. Confidence: Medium. Justification: The debts of the company are increasing. This is a result of the new investments of the company. As we are in a high interest rate epoch, many investors are concerned about this debt level. Synopsys has boosted its EBIT by 33%, thus it can amenize the situation of the future debt repayments. The company has net cash of US\$1.98b. Despite these positive numbers, still the cost of capital is high and the company is not profitable enough.
(<https://simplywall.st/stocks/us/software/nasdaq-snps/synopsys/news/is-synopsys-nasdaq-snps-using-too-much-debt-2>,
<https://www.design-reuse.com/news/57149/synopsys-2024-financial-results.html>)
- Zscaler (ZS) – Absolute view. Q: -10%. Confidence: Low. Justification: Zero-trust security vendor Zscaler is overwhelmed in 2024 and the company is not profitable enough. The competitors are strong and maybe the company will not be able to catch up with them.
(<https://www.investors.com/news/technology/zscaler-stock-zscaler-earnings-q32024/>)
- Apple (AAPL) – Absolute view. Q: +8%. Confidence: Medium. Justification: Apple's stock is expected to grind higher at a modest rate. FY2024–25 consensus calls for single-digit growth. Without a major new product until Vision Pro (late 2024), Apple lacks catalysts – reflected in its ~ 28x P/E limiting upside. Thus, we expect a modest

positive excess return.
(<https://www.nasdaq.com/articles/will-microsoft-still-be-worth-more-than-apple-by-2025>)

- Microsoft (MSFT) – Absolute view. Q: +15%. Confidence: High. Justification: Microsoft is having good results in different sectors. Analysts project 15% EPS growth in FY2024 and FY2025. The stock’s valuation ($\approx 35\times$) is higher, but deemed justified by its stronger growth. Overall confidence is high due to MSFT’s leadership in AI and diversified revenue streams such as gaming and cloud computing.
(<https://www.nasdaq.com/articles/will-microsoft-still-be-worth-more-than-apple-by-2025>)

Consumer Staples

- Procter & Gamble (PG) – Absolute view. Q: +10%. Confidence: Medium. Justification: P&G is a defensive stalwart with modest growth. With steady consumer demand and easing commodity costs, P&G should deliver a roughly high-single-digit return, in line with its earnings growth plus dividend.
(<https://www.investing.com/news/insider-trading-news/procter--gambles-chief-hr-officer-sells-22-million-in-stock-93CH-3739659>)
- Kimberly-Clark (KMB) – Absolute view. Q: $\sim +5\%$. Confidence: Medium. Justification: The maker of Huggies and Kleenex is viewed as fairly valued. Consensus 12-month target $\sim 3.5\%$ dividend and low-single-digit sales growth yielding modest total return. In short, we expect a stable but unexciting outlook.
(<https://www.marketbeat.com/originals/dividend-king-kimberly-clark-is-a-good-buy-in-2025/>)
- Molson Coors (TAP) – Absolute view. Q: $\sim +5\%$. Confidence: Medium. Justification: Analysts have mixed views (5 Buy, 6 Hold). That modest upside reflects stable beer demand but few catalysts.
(<https://www.nasdaq.com/articles/where-molson-coors-beverage-stands-analysts>)
- Costco (COST) – Absolute view. Q: $\sim +8\%$. Confidence: Medium. Justification: Costco continues to be a steady “compounder.” Morgan Stanley called it “among the best” retailers
(<https://www.investopedia.com/what-you-need-to-know-ahead-of-costco-earnings-thursday-q4-fy-2024-earnings-preview-8715769>)

Utilities

- NextEra Energy (NEE) – Absolute view. Q: $\sim +15\%$. Confidence: Medium. Justification: After a 2024 sell-off (clean energy stocks hit by higher rates), NEE has significant

rebound potential.

(<https://www.nasdaq.com/articles/examining-future-nextera-energys-earnings-outlook>)

- Duke Energy (DUK) – Absolute view. Q: ~ +8%. Confidence: High. Justification: Duke is a regulated utility with a predictable outlook. The only issue that we see is the debt level of the company which concerns some investors. (<https://simplywall.st/stocks/us/utilities/nyse-duk/duke-energy/news/duke-energy-nyseduk-has-a-somewhat-strained-balance-sheet-2>)
- AES Corp (AES) – Absolute view. Q: ~ -10%. Confidence: Medium-Low. Justification: AES's stock decreased ~ -30% in 2024, but some analysts anticipate a partial rebound which we are not sure about. (<https://www.nasdaq.com/articles/aes-corporation-stock-outlook-wall-street-bullish-or-be-arish>)

Health Care

- -UnitedHealth Group (UNH) – Absolute view. Q: ~+10%. Confidence: Medium. Justification: UNH shares pulled back in 2024 on healthcare cost concerns, but the stock remains very bullish. (<https://www.unitedhealthgroup.com/content/dam/UHG/PDF/investors/2025/UNH-Reports-Q1-2025-Results-Revises-Full-Year-Guidance.pdf>)
- Medtronic (MDT) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: Medtronic is in recovery mode from the previous losses with new product launches. The profit is expected to grow by 15% in 2025 (<https://simplywall.st/stocks/us/healthcare/nyse-mdt/medtronic/news/medtronic-nysemdt-is-posting-promising-earnings-but-the-good>)
- Baxter International (BAX) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: Baxter underwent restructuring in 2024. Maybe we will see a rebound in 2025. There is a challenging outlook for all the sectors in 2025 due to the high cost of the health system. The company maintain a "good" financial health score (<https://www.investing.com/news/company-news/baxter-stock-touches-52week-low-at-316-amid-market-challenges-93CH-3764142>)
- Eli Lilly (LLY) – Absolute view. Q: ~+20%. Confidence: Medium. Justification: Lilly's new drugs for obesity and Alzheimer were very successful and help to increase gains in 2023. Many analysts see this company undervalued and expect a rebound in 2025. (<https://www.nasdaq.com/articles/eli-lilly-undervalued-growth-stock-buy-now>)

Consumer Discretionary

- Amazon.com (AMZN) – Absolute view. Q: ~+20%. Confidence: High. Justification: Amazon's cloud and retail businesses are reaccelerating after a cost efficiency drive. The

stock prices all near historic high. We don't see any reason for not expect continue this trend in 2025.

(<https://www.ig.com/en/news-and-trade-ideas/amazon-share-price-analysis--what-lies-ahead-for-the-tech-giant--241126>)

- Nike (NKE) – Absolute view. Q: ~+5%. Confidence: Low. Justification: Nike surprised the market with a strong Q4 earnings report and guidance for Q1 2025. Nike's EPS of \$0.78 exceeded forecasts by approximately 20%. But there are many market challenges that affected negatively the revenue. So we see a mix signal. (<https://www.investing.com/news/transcripts/earnings-call-transcript-nike-q2-2024-beats-eps-forecasts-stock-rises-93CH-3783503>)
- Etsy (ETSY) – Absolute view. Q: ~-5%. Confidence: Low. Justification: Etsy is struggling with a post-pandemic crisis – directors informed that Q1 2025 gross merchandise sales will probably decline again like Q4 2024. (<https://finance.yahoo.com/news/why-etsy-inc-nasdaq-etsy-120237563.html>)
- Tesla (TSLA) – Absolute view. Q: ~-10%. Confidence: Medium. Justification: Tesla's CEO is actively participating in the political discussions and this is a risk for the company. New Chinese car makers are growing fast and the market share of the company is shrinking. The company is not profitable enough and the debt level is growing. (<https://www.morningstar.co.uk/uk/news/258640/is-tesla-stock-a-buy-or-sell-after-massive-post-election-surge-.aspx>)

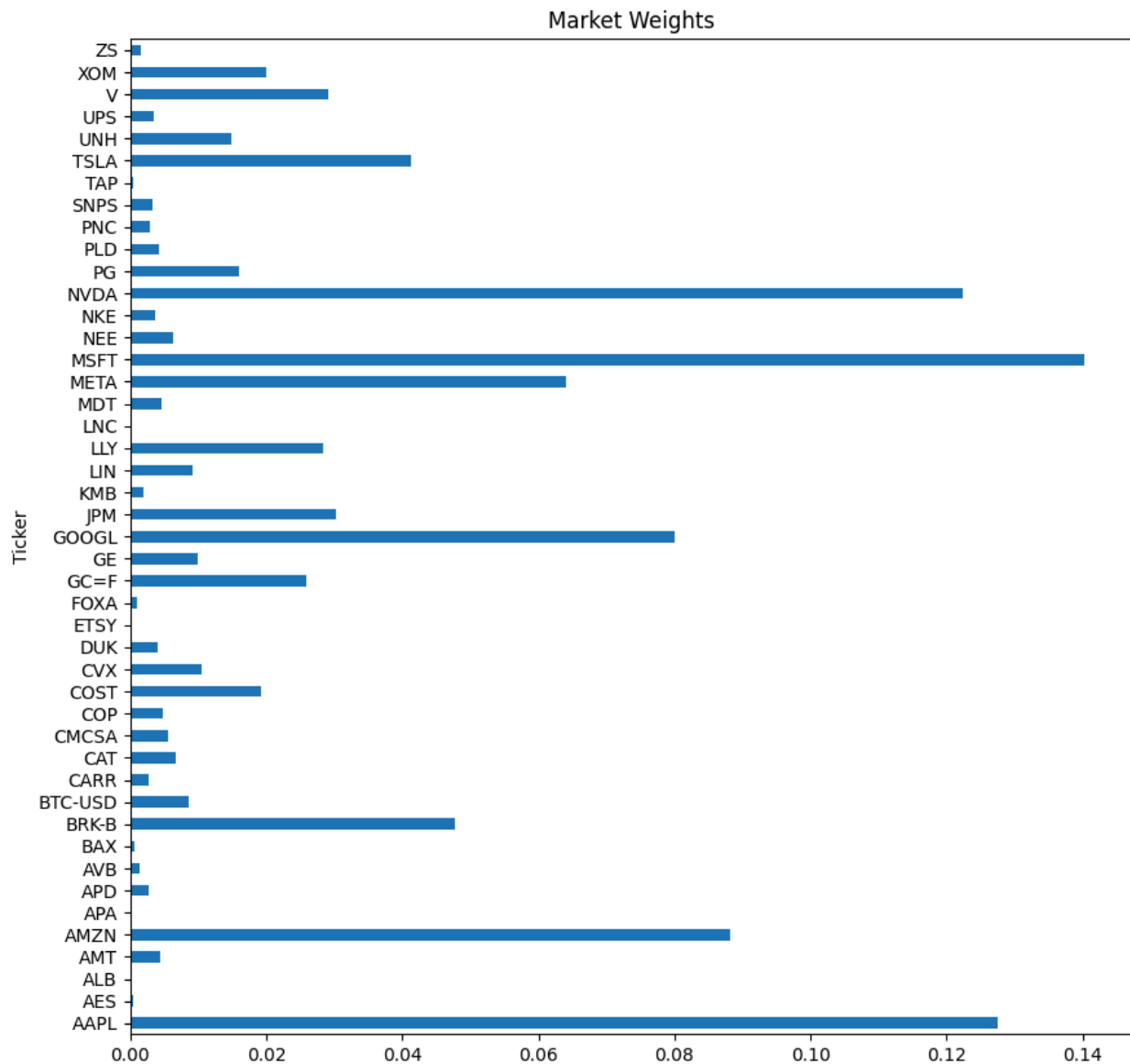
Real Estate

- Prologis (PLD) – Absolute view. Q: -5%. Confidence: Medium. Justification: Prologis, is the largest logistics REIT in the US market but due to the current challenges in the sector, it is expected to deliver modest gains. In fact, the PLD is underperforming the REIT sector compared to the rivals. (<https://www.nasdaq.com/articles/prologis-stock-pld-underperforming-real-estate-sector>)
- American Tower (AMT) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: Tower REITs, like PLD, are facing pressure from high rates and tenant mergers in 2024. But due to the surge of 5G technology and the necessity for new installations, the demand is still relatively strong for the company. (<https://finance.yahoo.com/news/american-tower-stock-falls-16-163900699.html>)
- AvalonBay Communities (AVB) – Absolute view. Q: ~+5%. Confidence: Medium. Justification: AvalonBay Communities is a multifamily residential REIT that specializes in upscale apartment communities. The same situation as the two previous companies here. The sector is struggling due to high interest rates and the demand for the apartments

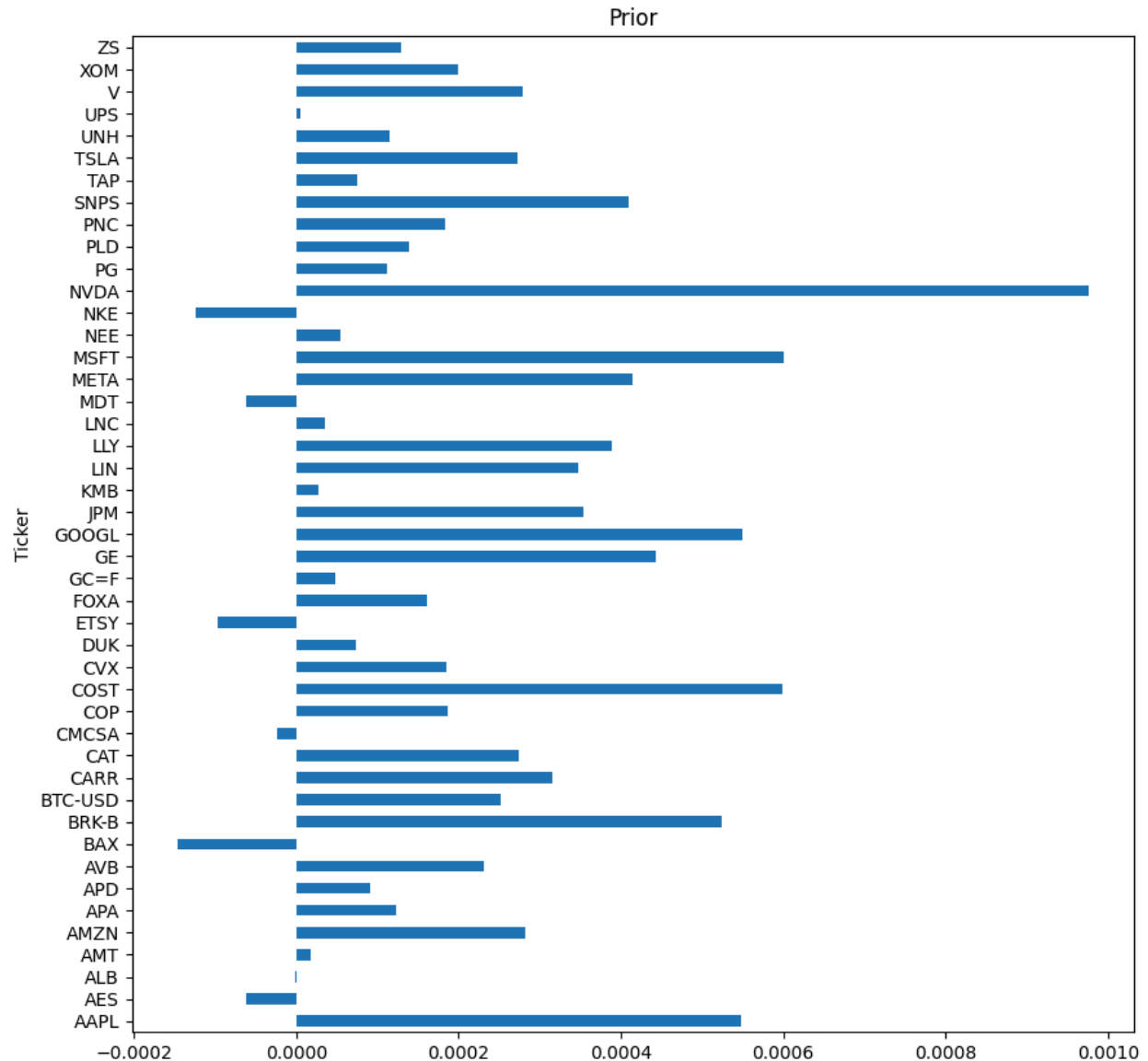
is decreasing.
(<https://money.usnews.com/investing/articles/best-real-estate-stocks-to-buy>)

The above views can be used to construct the P , Q and Ω matrix that reflects the opinions of the portfolio manager.

One of the main ingredients of the BL model is the market weight of each asset in our universe. We used the current market cap of the firms in the S&P 500 as a proxy for the weights on the final of 2024. For the market cap of Gold and Bitcoin we used some big but somehow arbitrary values. The result is as follow



Using this market cap and the δ we arrived at the below priors



Because Σw is the covariance of asset i with the cap-weighted market portfolio, the sign depends on the asset's beta:

- Positive beta $\rightarrow$ positive Π (asset expected to earn a positive excess return over the risk-free rate).
- Negative or very low beta $\rightarrow$ zero or negative Π (CAPM says it should earn less than the risk-free rate because it hedges the market).

So negative securities are not an error; they simply indicate assets that reduce portfolio volatility (low/negative beta) and therefore carry a lower equilibrium reward.

GROUP WORK PROJECT # 2
Group Number: 9284

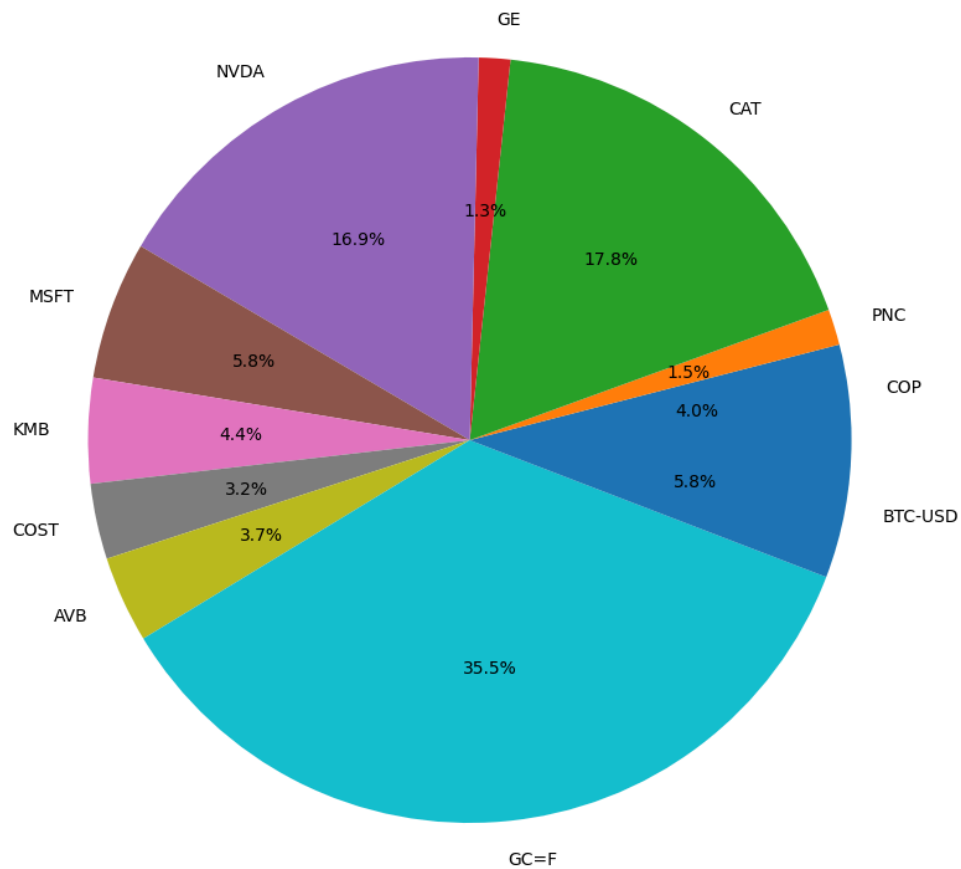
MScFE 652: PORTFOLIO MANAGEMENT

Let's evaluate the performance of the portfolio constructed by means of Black-Litterman model and compare it with the previous ones. Before the comparison we prone the weights less than 1% in the BL model to have better visualizations.

	MVO_Σsample	MVO_Σshrink	BL
COP	0.0	0.0	0.0402
PNC	0.0	0.0	0.0151
V	0.107	0.109	0.0
CAT	0.0	0.0	0.1776
GE	0.0	0.0	0.0133
NVDA	0.103	0.097	0.1691
SNPS	0.079	0.082	0.0
MSFT	0.0	0.0	0.0584
KMB	0.347	0.346	0.0444
COST	0.0	0.0	0.0319
MDT	0.122	0.124	0.0
AVB	0.0	0.0	0.0366
GC=F	0.243	0.242	0.3551
BTC-USD	0.0	0.0	0.0582

BL has given max weights to GC=F followed by CAT, NVDA,BTC-USD, MSFT, KMB whereas MVO has given max weights to KMB, followed by GC=F, NVDA,V and MDT. The weights of composing assets in the portfolio obtained by the BL model are

Portfolio Weights (Black-Litterman Covariance)



Contribution to portfolio variance (train Σ): (below)

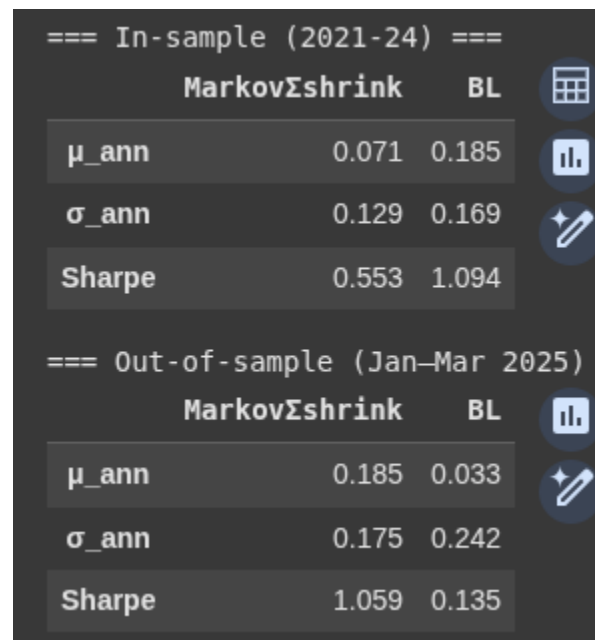
	MarkovΣshrink	BL
AVB	0.0	1.9425
BTC-USD	0.0	11.2078
CAT	0.0	18.8715
COP	0.0	3.0573
COST	0.0	2.1356
GC=F	9.4533	12.8767
GE	0.0	1.1947
KMB	28.6554	0.4447
MDT	11.0837	0.0
MSFT	0.0	5.7416
NVDA	25.1443	41.3959
PNC	0.0	1.1316
SNPS	14.1591	0.0
V	11.5042	0.0

Highest contribution to variance by MVO is by KMB, NVDA, SNPS, V.MDT and GC=F.

And finally the most important part, the performance of the models.

Highest contribution to variance for BL model is by NVDA, followed by CAT, BTC-USD and GC=F.

Risk-return tradeoff: Well, it is said that for higher returns one needs to take higher risks..



The image shows a screenshot of a software interface displaying performance metrics for two models: MarkovΣshrink and BL. The metrics are categorized into 'In-sample (2021-24)' and 'Out-of-sample (Jan-Mar 2025)'. The metrics include annualized return (μ_{ann}), annualized standard deviation (σ_{ann}), and the Sharpe ratio. The BL model shows significantly higher Sharpe ratios in both periods compared to the MarkovΣshrink model.

	MarkovΣshrink	BL
=== In-sample (2021-24) ===		
μ_{ann}	0.071	0.185
σ_{ann}	0.129	0.169
Sharpe	0.553	1.094
=== Out-of-sample (Jan-Mar 2025) ===		
μ_{ann}	0.185	0.033
σ_{ann}	0.175	0.242
Sharpe	1.059	0.135

The performance of our portfolio is not as expected. In fact, since the beginning of the year 2025 due to the tariff policies of the US government, the financial market has been very volatile and unstable. Somehow, we are experiencing a regime shift in the market. So, the importance of the historical data on the future performance of the portfolio is reduced.

4.3 Background on Black Litterman model and implementation

The Black-Litterman (BL) model is a Bayesian asset allocation framework for a portfolio that combines market equilibrium returns with investor specific views. It was introduced in 1992 to overcome limitations of traditional mean-variance optimization (MVO), such as extreme and unstable weights due to input sensitivity [1].

1. Market Equilibrium Prior: The Implied Returns

The implementation of BL model starts with getting the market-implied equilibrium returns, derived from the Capital Asset Pricing Model (CAPM):

$$\pi = \delta \cdot \Sigma \cdot w_m$$

$$\delta = E[R_m] - r_f \sigma_m^2$$

where:

- π : Implied excess returns (prior)
- δ : Risk aversion coefficient, estimated from historical market data
- Σ : Covariance matrix of returns (we used a shrinkage estimator)
- w_m : Market capitalization weights (current market portfolio)

2. Absolute views and Relative views as Investor Views

Investor views are encoded using a set of equations that expresses beliefs about future asset returns:

$$P\mu = Q + \epsilon$$

where:

- $P \in \mathbb{R}^{k \times n}$: The P matrix representing how each view relates to the assets.
- $Q \in \mathbb{R}^k$: Expected return of each view (can be absolute or relative).
- ϵ : A normally distributed error term with covariance Ω (Omega) $\sim N(0, \Omega)$: represents the error or uncertainty in each view

Absolute views correspond to investors beliefs about an asset's expected return (e.g., $E[r_i] = q$ or Asset A will return 5%), while relative views express expectations about differences in performance (e.g., $E[r_i - r_j] = q$ or Asset A will outperform Asset B by 2%).).

The matrix Ω (Omega) plays an important role, as it determines how much influence each view has on the final posterior. A more confident view (smaller uncertainty) will shift the posterior returns closer to the view.

$$\Omega = \text{diag}(\tau P \Sigma P^T / \text{confidence_scale}^2)$$

where:

- τ (tau): A small scalar (often between 0.025 and 0.05) reflecting the uncertainty in the prior.
- $P \Sigma P^T$: The variance of the portfolio under the prior covariance.
- confidence_scale : A scalar from a predefined scale: e.g.

High confidence: small denominator (e.g., 0.1)

Medium confidence: moderate (e.g., 0.3)

Low confidence: higher uncertainty (e.g., 1.0)

This formulation will ensure that each view's uncertainty is properly adjusted for both its internal volatility and the investor's stated confidence level

3. Posterior Distribution

The BL model updates the prior mean π (π) and covariance Σ (Sigma) using Bayesian inference to obtain the posterior estimates:

$$M_{BL} = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}Q]$$

$$\Sigma_{BL} = \Sigma + [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}$$

These adjusted posterior moments reflect both market equilibrium and investor views.

4. Optimization and Post-processing

The posterior μ_{BL} and Σ_{BL} are given as inputs to our MVO weights function :

$$\text{Max}_w (w^T\mu_{BL} - \lambda/2 w^T\Sigma_{BL}w \text{ subjected to: } w \in [w_{\min}, w_{\max}], \text{ and } \sum w = 1$$

After solving for the optimal weights, we apply a pruning function:

`prune_weights(w, floor=0.01)`

- This eliminates negligible weights (less than 1%) to ensure the final portfolio is implementable and sparse

5. Interpretation

- π (π) represents the best guess from the market without any subjective inputs.
- Views allow overlaying investor beliefs in a structured, quantitative way.
- Ω (Omega) captures uncertainty in views, allowing flexible expression of confidence.
- Posterior returns and covariance reflect a blend of market equilibrium and adjusted beliefs.

6. Implementation Notes

- We have used shrinkage estimators for Σ (Sigma) to improve numerical stability.

- Market weights are obtained using data from yahoo finance) and adjusted manually for non-equity assets like gold and Bitcoin.
- The final output compares the BL weights with traditional MVO solutions.
- Below Inputs and outputs table:

Element	Symbol	Description
Prior mean	Π (pi)	Implied returns from CAPM
Prior cov.	Σ (Sigma)	Shrinkage estimator
View matrix	P	Which assets each view targets
View returns	Q	Expected returns from views
View variance	Ω (Omega)	Uncertainty of views
Posterior mu	μ_{BL}	Adjusted expected returns
Posterior cov	Σ_{BL}	Adjusted covariance matrix

Step 5: Kelly Criterion

1. The Kelly Formula

For asset with expected return μ , variance σ^2 , and risk-free rate r_f , the Kelly fraction f^* is given by [2]:

$$f^* = (\mu - r_f) / \sigma^2$$

where:

μ : Expected return of the asset (usually daily or annualized)

r_f : Risk-free rate (e.g., daily treasury yield)

σ^2 : Variance of the asset's returns

f^* : Optimal fraction of portfolio to allocate to that asset

This maximizes expected logarithmic growth of wealth:

$$\max_f E[\log(1 + fR)]$$

2. For Multi Asset Portfolios

In a portfolio context, the Kelly rule becomes:

$$w^* = \Sigma^{-1} (\mu - r_f)$$

where:

μ : Vector of expected excess returns

Σ : Covariance matrix of returns

w : Portfolio weights

3. Inputs

We have already expected return | μ : historical mean returns (daily), Risk-free rate, Covariance, Σ ,
Sample or shrinkage estimator |

4. Backtesting Procedure

1. Estimate kelly weights per asset using in-sample returns_train data.
2. Apply weights to test data and calculate portfolio growth i.e. cumulative returns:

$$\text{Port_val_t} = \text{Prt_val_}\{t-1\} (1 + w^T r_t)$$
3. Compare kelly returns against:
 - Equal-weight (1/N)
 - Mean-variance (MVO)
 - Black-Litterman (BL)
4. Evaluate final wealth, Sharpe ratio, volatility.

5. Interpretation and Future Considerations

Full Kelly is aggressive; fractional Kelly is common in practice.
Results are sensitive to input estimation.

Kelly adjusts for diversification and covariance like other MVO and BL models.
Weekly/monthly rebalancing is recommended for stability.

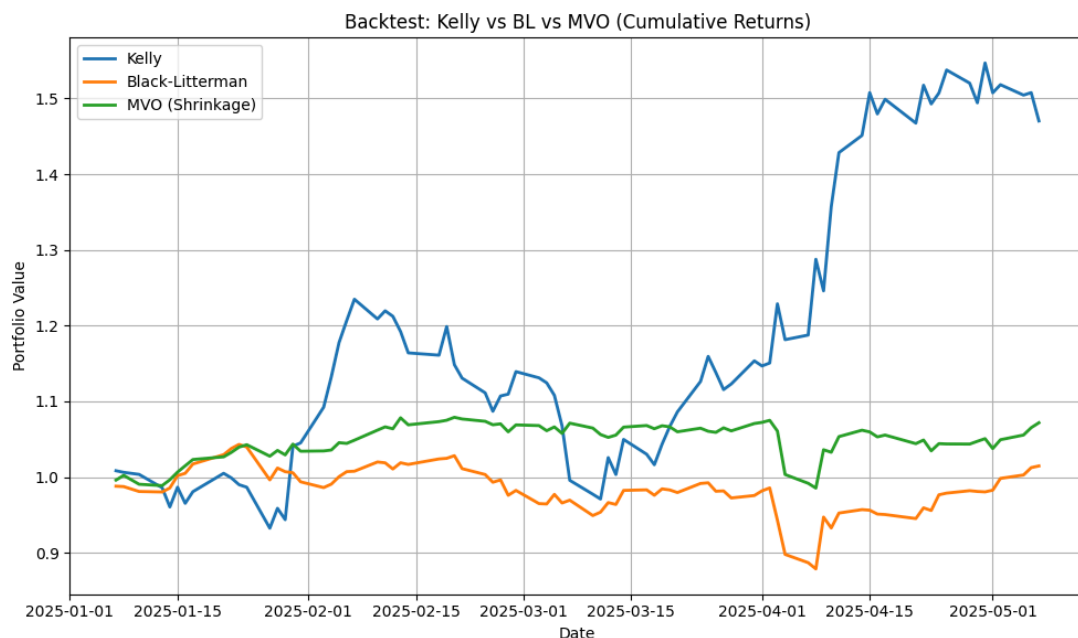
6. Kelly Results and Comparison

Kelly offers optimal growth when calibrated correctly.

Kelly Performance:

```
Ann. Return      1.2908
Ann. Volatility   0.4927
Sharpe Ratio     2.5576
Total Return     0.4703
dtype: float64
```

Better Sharpe Ratio and better growth than MVO and BL



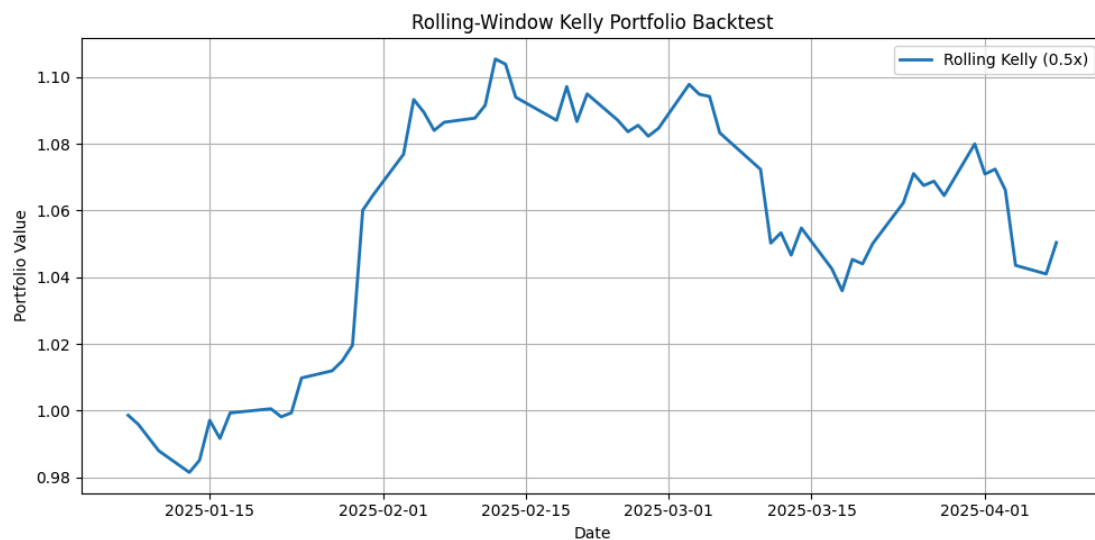
7. Rolling-Window Kelly Backtest (with rebalancing)

We can use fractional Kelly (e.g., 0.5x) for reduced volatility
Rolling-window estimation for dynamic rebalancing

Steps:

- Re-estimate fractional (0.5) Kelly weights every N days (e.g., 21)

- Applies weights to the next N days
- Accumulates portfolio value over time



Final Notes:

- BL: combines market belief and investor views
- We compared performance to MVO and Kelly under test conditions.
- BL has given max weights to GC=F followed by CAT, NVDA, BTC-USD, MSFT, KMB whereas MVO has given max weights to KMB, followed by GC=F, NVDA, V and MDT.
- Highest contribution to variance for BL model is by NVDA, followed by CAT, BTC-USD and GC=F.
- Risk-return tradeoff: Well, it is said for higher returns one need to take higher risks..
- Highest contribution to variance for MVO is by KMB, NVDA, SNPS, V.MDT and GC=F
- In long run Kelly outperformed MVO and BL because of many assets going short and ability to fractional kelly and monthly rebalancing
- High volatility do impact the prediction ability

References:

[1] Meucci, Attilio. "The Black-Litterman Approach: Original Model and Extensions." *SSRN Electronic Journal*, 1 Aug. 2008, <https://doi.org/10.2139/ssrn.1117574>.

[2] Carta, Andrea, and Claudio Conversano. "Practical Implementation of the Kelly Criterion: Optimal Growth Rate, Number of Trades, and Rebalancing Frequency for Equity Portfolios." *Frontiers in Applied Mathematics and Statistics*, vol. 6, 2020, article 577050. <https://doi.org/10.3389/fams.2020.577050>.